

User Manual: setting up and reading your sensor unit

Daffodil watches water for you. It reads water troughs, tanks, septic tanks, water flow and water temperature, then reports to Annabelle, the farm's base station, over LoRa radio and to Digital Stables over WiFi. It runs on solar power and a battery.

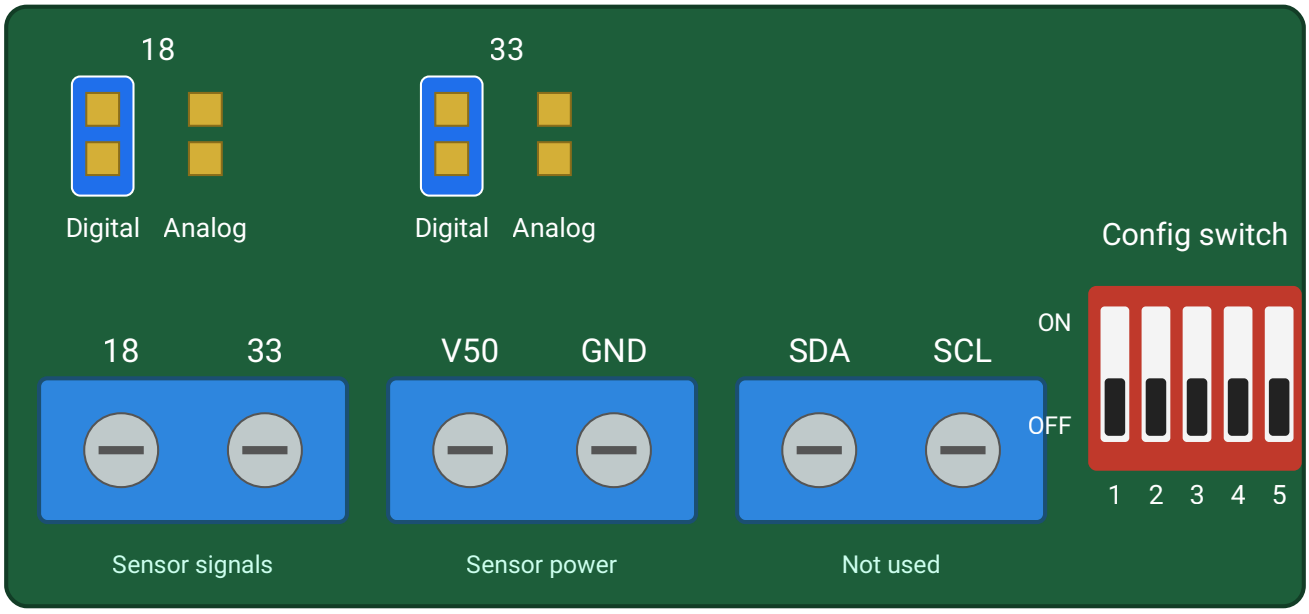
Every chapter has two parts. Basics is everything you need to set up and use Daffodil. Details, in the grey part at the end of each chapter, explains how it works, with the numbers and formulas, for anyone who wants to know what is going on inside.

Digital Stables · October 2026

1. Getting to know your Daffodil

BASICS

Everything you need to set up is on the lower edge of the board:



Part	What it does
18 / 33 terminal	The signal wires of your sensors go here. Terminal 18 is sensor 1 and terminal 33 is sensor 2.
V50 / GND terminal	Powers your sensors. Every sensor's + wire goes to V50 and its − wire goes to GND. When two sensors are connected, both + wires share V50 and both − wires share GND.
18 and 33 jumpers (marked "Digital Analog")	Each sensor terminal has two small 2-pin headers, one labelled Digital and one labelled Analog. Put one jumper cap on the header your sensor needs and leave the other one empty. The drawing above shows both caps on Digital.
Config switch (1–5)	Tells Daffodil what it is measuring (switches 1–4) and how often to report (switch 5). Up is ON.
SDA / SCL terminal	Not needed for any of the setups in this manual.

> Which header? Use Digital for flow meters, ultrasonic level sensors and the water-temperature probe. Use Analog only for tank pressure sensors. Never put a cap on both headers of the same terminal.

> Always switch the unit off (unplug USB and the battery) before changing wires, jumpers or switches. Daffodil only reads the config switch when it starts up.

DETAILS

What is inside the box, and how the power flows.

Daffodil is two boards stacked together: the Wally base board, which carries the ESP32 processor, the LoRa radio (433 MHz), WiFi, the clock and the current sensors, and the Daffodil shield, which carries the terminals, the config switch and the battery charger.

Part	Role
Solar panel → charger → battery	The panel charges a single LiFePO4 cell, 600 mAh (3.2 V nominal, 3.6 V when full). The charger stops at 3.6 V.
Boost converter	Makes the 5 V supply for the sensors (V50) from the battery whenever the panel can't. In full sun the panel supplies it directly.
Two current sensors	One on the panel, one on the battery (50 mΩ shunts). Each reading is the average of 128 samples, because the charger switches on and off many times a second and single snapshots would be misleading.
Processor (ESP32, 2 cores)	Runs at 80 MHz to save power. One core runs the program; the other counts the charge going in and out of the battery (chapter 10).
Real-time clock	Keeps time while the processor sleeps. Corrected from Annabelle's forecast broadcasts (chapter 7).
Watchdog (TPL5010)	A separate chip that restarts the processor if it ever stops responding (chapter 8).

2. Quick setup table

BASICS

Find what you want to measure, then set the switches, sensors and jumper caps exactly as shown. In the switch pictures, a black knob at the top means ON. Switch 5, shown with a ?, is your choice (chapter 3).

Switches 1–4	Setting	Terminal 18	18 jumper	Terminal 33	33 jumper
 off off off off	1 Flow Sensor Measures water flow through one pipe.	Flow meter	 Digital	nothing	 No cap
 ON off off off	2 Flow Sensors Measures flow through two pipes.	Flow meter 1	 Digital	Flow meter 2	 Digital
 off ON off off	1 Flow Sensor + 1 Tank One pipe's flow plus one tank's level.	Flow meter	 Digital	Tank pressure sensor	 Analog
 ON ON off off	1 Tank Level of one tank, using a pressure sensor at the bottom.	Tank pressure sensor	 Analog	nothing	 No cap
 off off ON off	2 Tanks Levels of two tanks.	Tank 1 pressure sensor	 Analog	Tank 2 pressure sensor	 Analog
 ON off ON off	Septic Tank How full a septic tank is, measured from above.	Ultrasonic sensor	 Digital	nothing	 No cap
 off ON ON off	Water Trough Water level in a trough, measured from above.	Ultrasonic sensor	 Digital	nothing	 No cap
 ON ON ON off	Water Trough + Tank A trough level plus one tank level. The ultrasonic goes on terminal 33 in this setting.	Tank pressure sensor	 Analog	Ultrasonic sensor	 Digital
 off off ON ON	2 Water Troughs Water levels in two troughs. Both levels are reported, but the lights don't show them yet.	Ultrasonic sensor 1	 Digital	Ultrasonic sensor 2	 Digital
 off off off ON	Water Trough + Water Temperature A trough level plus the water temperature.	Ultrasonic sensor	 Digital	Water temperature probe	 Digital

Jumper pictures: D = Digital header, A = Analog header. The blue block shows where the cap goes. Any switch combination not in this table is unused, and the unit won't measure anything in it.

DETAILS

How Daffodil reads five switches through one wire.

The five switches form a resistor ladder: each switch adds a different resistor, so every one of the 32 positions produces a different voltage. Daffodil reads that one voltage with its analog converter (averaged over 8 readings) only when it starts up, and looks it up in a table to find the setting and whether switch 5 is on.

The ladder is powered from V50, which varies a little from one unit and battery to another. So each unit is calibrated once at installation: with all switches off, it stores its own reading, and every later reading is scaled by `factor = 8326 ÷ stored reading` before the lookup. A new battery can shift the readings, which is why chapter 8 says to recalibrate after a battery change.

3. Switch 5: how often to report

BASICS

Switch 5 ON: Battery Saver

Daffodil plans its day around the sun. It sleeps longer and reports less often, so the battery lasts as long as possible. Best for most installations, such as a trough level that only needs checking a few times an hour.

Switch 5 OFF: Frequent Reports

Daffodil reports more often and doesn't wait for good sun. You get readings closer to real time, but it uses more battery. Best when you need to watch something closely, such as water flow, or when the unit has reliable power.

Either way, Daffodil protects its own battery. If the battery gets low it turns off its lights first, then WiFi, then goes to sleep until the sun has recharged it.

At night Daffodil still reports at least every 5 minutes while its battery is healthy, and every 90 seconds in the hour and a half before sunrise, to catch the coldest part of the night. If the battery runs low it sleeps longer between reports to save it.

DETAILS

When Daffodil sleeps, for how long, and the battery protection steps.

Battery Saver (switch 5 ON)

Daffodil works out from the date, time and its latitude how strong the sun should be right now (the solar efficiency, 0–100%). It stays awake, with lights on, while the efficiency is above the unit's minimum for lights (40% by default), and starts WiFi above the WiFi minimum (50%). Below that it sleeps between reports, unless the hardware says the sun really is working: panel current at least 30 mA, the panel-side supply (V50_I) at least 3.5 V, or the battery charging. When the sun model says "sunny" but the panel current is under 30 mA or V50_I is under 3.5 V, the unit marks itself cloudy and also sleeps between reports.

How long it sleeps

```
Daytime (sun up):            60 s
Night:   sleep = 450 × night length (min) ÷ minutes to sunrise   (at least 90 s)
Last 90 min before sunrise:  90 s
Night cap: never more than 300 s while the battery is at or above 3.15 V
```

Example, 11-hour night, sunrise 6:30	Formula	Battery ≥ 3.15 V	Battery < 3.15 V
8:00 pm (630 min to sunrise)	450 × 660 ÷ 630 = 471 s	300 s (5 min)	471 s (8 min)
2:00 am (270 min)	450 × 660 ÷ 270 = 1100 s	300 s (5 min)	1100 s (18 min)
5:30 am (60 min)	pre-dawn	90 s	90 s

Battery protection, in order

Battery voltage	What Daffodil does
below 3.28 V	WiFi off (it needs 3.35 V held for 30 s to start again)
below 3.18 V	Lights off
below 3.12 V	Forced sleep between reports, whatever switch 5 says
below 2.80 V	Rest mode: no measuring, one short "I am resting" message, then it only checks the battery until the sun brings it back. It keeps a log of the voltages while resting and sends it once it recovers.

4. Step-by-step setup

BASICS

1. Unplug the battery and USB.
2. Set switches 1–4 for what you're measuring (chapter 2), and switch 5 for how often it reports.
3. Fit the jumper caps for terminals 18 and 33 as shown in chapter 2.
4. Connect each sensor: its signal wire to terminal 18 or 33, its + wire to V50 and its − wire to GND (chapter 5).
5. Connect the battery (and USB or solar panel). The lights start cycling through their displays.
6. For trough and septic setups, enter the sensor's mounting height (chapter 6).
7. Watch the lights for a minute and check the level, internet and LoRa displays look right (chapter 7).

The setups one by one:



1 Flow Sensor

Measures water flow through one pipe.

- Terminal 18: Flow meter signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



2 Flow Sensors

Measures flow through two pipes.

- Terminal 18: Flow meter 1 signal wire. Cap on the 18 Digital header.
- Terminal 33: Flow meter 2 signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).



1 Flow Sensor + 1 Tank

One pipe's flow plus one tank's level.

- Terminal 18: Flow meter signal wire. Cap on the 18 Digital header.
- Terminal 33: Tank pressure sensor signal wire. Cap on the 33 Analog header.
- Every sensor's power wires: V50 (+) and GND (−).



1 Tank

Level of one tank, using a pressure sensor at the bottom.

- Terminal 18: Tank pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



2 Tanks

Levels of two tanks.

- Terminal 18: Tank 1 pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: Tank 2 pressure sensor signal wire. Cap on the 33 Analog header.
- Every sensor's power wires: V50 (+) and GND (−).



Septic Tank

How full a septic tank is, measured from above.

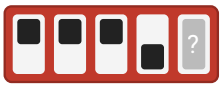
- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



Water Trough

Water level in a trough, measured from above.

- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: leave empty. No jumper cap needed.
- Every sensor's power wires: V50 (+) and GND (−).



Water Trough + Tank

A trough level plus one tank level.

- Terminal 18: Tank pressure sensor signal wire. Cap on the 18 Analog header.
- Terminal 33: Ultrasonic sensor signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

The ultrasonic goes on terminal 33 in this setting.



2 Water Troughs

Water levels in two troughs.

- Terminal 18: Ultrasonic sensor 1 signal wire. Cap on the 18 Digital header.
- Terminal 33: Ultrasonic sensor 2 signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

Both levels are reported, but the lights don't show them yet.



Water Trough + Water Temperature

A trough level plus the water temperature.

- Terminal 18: Ultrasonic sensor signal wire. Cap on the 18 Digital header.
- Terminal 33: Water temperature probe signal wire. Cap on the 33 Digital header.
- Every sensor's power wires: V50 (+) and GND (−).

DETAILS

What happens when Daffodil starts up.

1. Reads the config switch (chapter 2) and checks for a battery: below 1 V counts as "no battery", and then the battery protection of chapter 3 is switched off.
2. Starts its sensors, the clock and the radio.
3. After a real restart (power-on, brownout, crash or watchdog) it sends Annabelle two short messages: its identity (name, serial number, firmware) and its power profile (battery type and size, chapter 10). It repeats them once a day.
4. Decides whether to stay awake or sleep (chapter 3).

5. Connecting the sensors

BASICS

Ultrasonic level sensor (troughs and septic tanks)

Use a waterproof ultrasonic sensor with automatic serial (UART) output, such as the A02YYUW. JSN-SR04T and AJ-SR04M sensors also work if set to their automatic serial mode. The older style of sensor with separate "trigger" and "echo" wires and open speakers does not work with current Daffodils. Those sensors corroded in the field.

Sensor wire	Connect to
VCC / + (usually red)	V50
GND / − (usually black)	GND
TX	Terminal 18. In Water Trough + Tank, terminal 33. In 2 Water Troughs, the second sensor goes on 33.
RX	Not connected. Tape it off.

Jumper cap on that terminal's Digital header.

> Mounting: point the sensor straight down at the water, clear of the trough walls, float valves and pipes. Keep its face at least 3 cm above the highest water level, because it can't measure anything closer than that.

Water temperature probe

A waterproof DS18B20 probe, the common stainless-steel tube on a cable. It is used only in the Water Trough + Water Temperature setting.

Probe wire	Connect to
Red	V50
Black	GND
Yellow (data)	Terminal 33, jumper cap on the 33 Digital header

No extra resistor is needed, because the board already has one. Put the probe tip in the water, away from the trough's inlet, so it measures the water the animals are actually drinking.

Flow meter

A standard 3-wire hall-effect water flow meter.

Meter wire	Connect to
Red	V50
Black	GND
Yellow (pulse)	Terminal 18 (flow meter 1) or 33 (flow meter 2), cap on Digital

Install the meter with the arrow on its body pointing in the direction the water flows.

Tank pressure sensor

A submersible or bottom-mounted pressure sensor with a 0.5–4.5 V output (0–5 psi), placed at the bottom of the tank.

Sensor wire	Connect to
+ (usually red)	V50
− (usually black)	GND
Signal	Terminal 18 (tank 1) or 33 (tank 2), cap on Analog

DETAILS

How each sensor's signal becomes a reading.

Sensor	How it is read
Ultrasonic	The sensor sends a 4-byte frame at 9600 baud: a start byte (0xFF), the distance in mm as two bytes, and a checksum = (0xFF + high + low) & 0xFF. Frames with a wrong checksum are thrown away. Distances under 3 cm can't be measured.
Water temperature	DS18B20 on a one-wire bus, 9-bit resolution (0.5 °C steps).
Flow meter	Each turn of the meter's wheel sends pulses. `flow (L/min) = pulses per second ÷ Q-factor`. The Q-factor depends on the meter model (defaults 0.35 for meter 1, 0.82 for meter 2) and can be set per unit. Daffodil also keeps a running total in millilitres.
Tank pressure	Read by the analog converter. `pressure (psi) = (volts − 0.5) × 1.25`. 0 V means a broken wire, because the sensor gives 0.5 V at zero pressure. 1 psi is about 70.3 cm of water above the sensor.

6. Setting up a trough or septic tank

BASICS

For troughs, Daffodil needs to know how high the sensor is and where you want the "low" and "full" points. Open the Daffodil's web page, tap the gear icon on the level card, and enter:

Setting	What to enter
Sensor height	Distance in cm from the sensor's face down to the empty trough floor.
Level minimum	Water depth in cm below which the trough counts as low (red light).
Level maximum	Water depth in cm above which the trough counts as full (blue light).

Example: the sensor is 60 cm above the trough floor, you want an alert below 20 cm of water, and you call it full above 30 cm. Enter 60, 20 and 30. Re-enter these whenever you move the sensor or change the trough.

A septic tank needs no settings. Its lights show how full the tank is, measured from a sensor mounted at the top.

DETAILS

From a distance to a water depth.

```
water depth (cm) = sensor height − measured distance
depth < level minimum            → low   (red)
level minimum ≤ depth ≤ maximum  → good  (green)
depth > level maximum            → full  (blue)
```

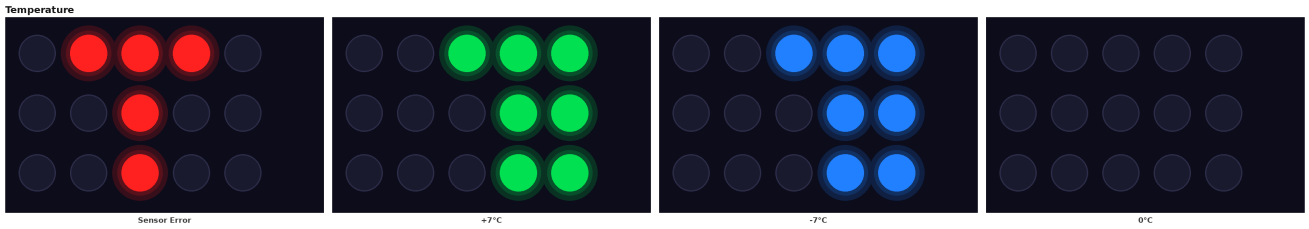
With the example above, a measured distance of 35 cm gives 60 − 35 = 25 cm of water: green. The settings are stored in the unit's memory and survive restarts.

7. Reading the lights

BASICS

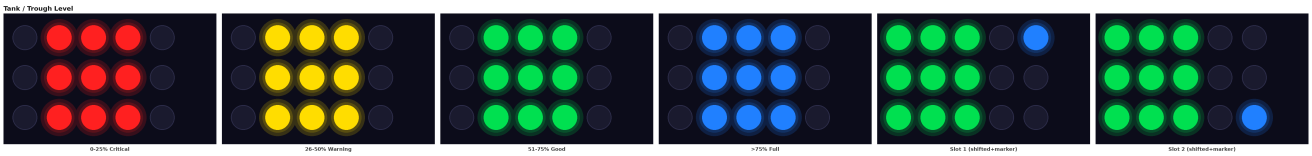
Daffodil has 15 lights in 3 rows of 5. Every few seconds the display moves on to show the next piece of information, in this order.

Temperature



The air temperature. The left lights count the tens and the right lights count the units. Green means above zero, blue means below zero, and yellow means exactly 0 °C. A red pattern means the temperature sensor couldn't be read.

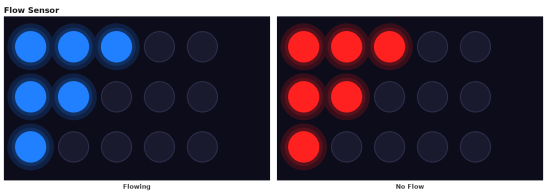
Water level (troughs, tanks, septic)



Red = low or critical, yellow = getting low (tanks and septic only), green = good, blue = full. For troughs, red, green and blue follow the levels you set in chapter 6. With two sensors, the block shifts left and a small blue dot shows which one you're seeing: top-right for sensor 1, bottom-right for sensor 2.

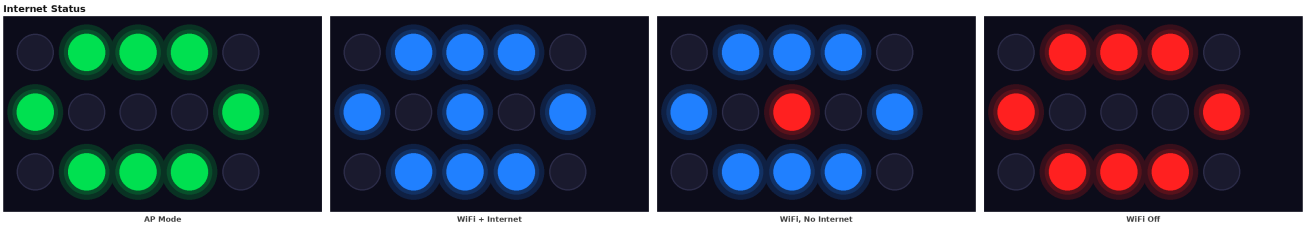
> A trough that always shows blue may mean the sensor isn't being heard at all. Check its wiring (chapter 9).

Water flow



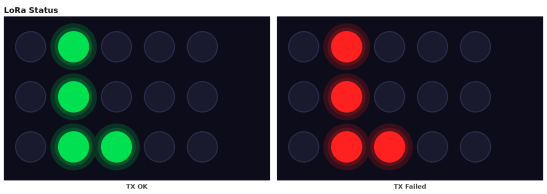
An "F" shape. Blue = water is flowing, red = no flow right now.

Internet



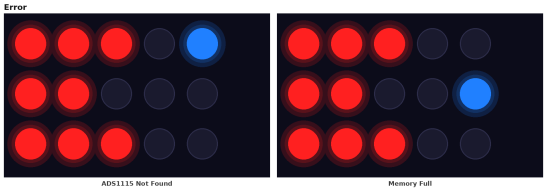
Blue = connected to your WiFi. The centre light is blue when the internet can be reached and red when it can't. Green = setup mode: Daffodil is broadcasting its own WiFi network for you to connect to. All red = WiFi is resting to save battery.

LoRa radio



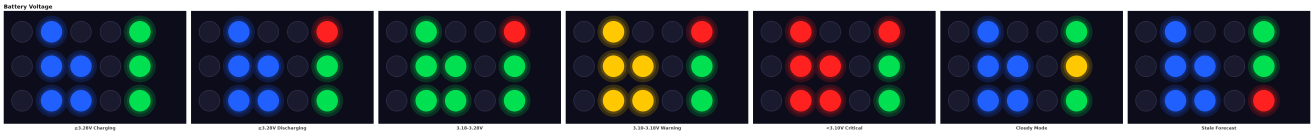
Green = the last radio message went out, red = it failed. The lights blink off briefly each time a message is sent. That's normal.

Error



A red "E" with a blue dot. A dot top-right means an internal sensor wasn't found. A dot in the middle-right means the unit's memory is nearly full. Either way, contact Digital Stables.

Battery



The "B" shows battery health: blue = good, green = OK, amber = low, red = very low, magenta = no battery detected. The top-right light is green while charging and red while running on battery. The middle-right light is amber on cloudy days, when the unit is saving power. The bottom-right light is green while Daffodil has a recent weather forecast from Annabelle and red when it doesn't.

DETAILS

The exact thresholds behind the battery display.

Light	Colour and meaning
"B" (battery voltage)	blue ≥ 3.28 V (WiFi allowed) · green 3.18–3.28 V · amber 3.10–3.18 V · red 1.0–3.10 V · magenta under 1.0 V (no battery)
Top-right (power source)	green battery current below −1 mA (charging) · red above +1 mA (running on battery) · blue in between
Middle-right (mode)	green full power mode · amber cloudy (chapter 3) · off otherwise
Bottom-right (weather)	green when the last forecast accepted from Annabelle is less than 31 minutes old · red otherwise

Weather forecasts and the clock

Annabelle broadcasts a weather forecast every 5 minutes, exactly on :00, :05, :10 and so on. The broadcast also carries Annabelle's time and a security code made from a secret the two share. Daffodil accepts it only if the code matches and the time is later than the last one it accepted; it then sets its own clock from it if it is off by more than a few seconds. While Daffodil sleeps it isn't listening, so a sleeping unit rarely catches a broadcast and its weather light is usually red at night. That is expected.

8. Looking after your Daffodil

BASICS

- Keep the solar panel clean and in full sun.
- Every few months, check the ultrasonic sensor's face is clean and the cable glands are tight.
- If you change the battery, the config switch may need recalibrating. Use Config Switch Calibration on the web page, or ask Digital Stables.
- If you move a sensor or change a trough, re-enter its heights (chapter 6).
- After a firmware update over USB, move the jumper from the USB position back to the watchdog header. Left on USB, the USB chip stays powered and can flatten the battery overnight. Annabelle's screen then shows !! USB JUMPER ON !!, and on the Teleonome page the unit's name turns red with USB header connected (see the phone picture in chapter 10).

DETAILS

The watchdog, the USB jumper and restarts.

Watchdog

The TPL5010 watchdog sends Daffodil a "wake" pulse about every 10 minutes, and Daffodil must answer it. If no answer comes for about 20 minutes, the watchdog restarts the processor. The pulse also wakes Daffodil from sleep, so a unit sleeping longer than 10 minutes wakes briefly, answers and goes back to sleep: these show as Early Wakes in Vital Signs and are normal. With the 5-minute night sleeps the timer always wakes first.

USB jumper

In the USB position the jumper powers the USB chip (10–20 mA, all the time). That is 240–480 mAh a day, so it can empty the 600 mAh battery in a day or two. Daffodil senses the jumper (its MODE pin reads above 1 V) and reports it, which is what turns the unit's name red on the Teleonome page.

Restarts

Reset reason	Meaning
POWERON	Power was connected, or a firmware update or the watchdog restarted it. Normal after an update or a battery change.
BROWNOUT	The voltage dipped too low for the processor, usually a weak or nearly empty battery under load.
PANIC, WDT	A software crash or a stuck program. If it repeats, contact Digital Stables.

9. Troubleshooting

BASICS

Problem	What to check
The level never changes, or a trough always shows blue	The ultrasonic's TX wire is on the right terminal (chapter 2) and its cap is on Digital. It has power from V50/GND, and it's an automatic serial sensor.
Water temperature missing	The probe's yellow wire is on terminal 33 with the cap on 33 Digital, and switches 1–4 are set to Water Trough + Water Temperature.
Flow always shows red (no flow)	The meter's yellow wire is on the right terminal with its cap on Digital, and the meter's arrow points the way the water flows.
Tank level wrong or stuck	The cap for that terminal is on Analog, not Digital.
The unit measures the wrong thing	Switches 1–4 are set correctly and the unit was restarted after changing them. If they are, the switch needs recalibrating (chapter 8).
Internet lights all red	Normal when the battery is low or it's very cloudy. Daffodil rests WiFi to save power and tries again later.
The unit's name is red with USB header connected	The programming jumper was left on the USB position after an update. Move it back to the watchdog header (chapter 8).
An orange or red Reset … ago tag on the Teleonome page	The unit restarted. Orange (power-on, shown for 1 hour) is normal after an update, a replug or a battery change. Red (brownout, crash or watchdog, shown for 24 hours) means something went wrong: if it keeps coming back, contact Digital Stables.
A % delivered tag, or readings arriving with long gaps	Some radio messages are not reaching Annabelle. Check the antenna is screwed on and upright, and that nothing new (a shed, a vehicle, a metal tank) stands between the unit and Annabelle.
The two battery % figures disagree	A few percent apart is normal (chapter 10). If they differ by more than about 15% for days, tell Digital Stables.
No lights, or very dim lights	Normal at night (the lights dim), when the battery is low, and on dull days in Battery Saver mode (the lights turn off to save power). Wait for sun, or check the battery is connected.

Still stuck? Contact Digital Stables with the unit's name and a description of what the lights are doing.

DETAILS

Where to look on the phone, and what the numbers mean.

Open the unit's window on the Teleonome page and go to Status › Diagnostics › Vital Signs. Daffodil sends these with every report.

Vital Sign	What it tells you
Last Reset Reason / Time, Reset Count	When and why it last restarted (chapter 8). A count that keeps climbing means repeated restarts.
Wakes, Early Wakes, Aborted Wakes	How often it woke; early wakes are watchdog pulses (normal); aborted wakes mean the battery was too low to do any work.
Awake %	Share of the time spent awake. High on sunny days, a few percent at night.
Wake Voltage, Battery Sag, Lowest TX Voltage	The battery's voltage at waking, how far it dips under load, and its lowest point while the radio sends. A big sag points to a tired battery.
LoRa TX Fail Count	Messages the radio itself failed to send.
Records Received / Missed, Delivery %	Every Vital Signs message carries a number that goes up by one each time. The Teleonome counts the gaps in those numbers since the last restart: `delivery % = received ÷ (received + missed)`
Battery Ledger, Power Profile	See chapter 10.

10. How Daffodil knows how full its battery is

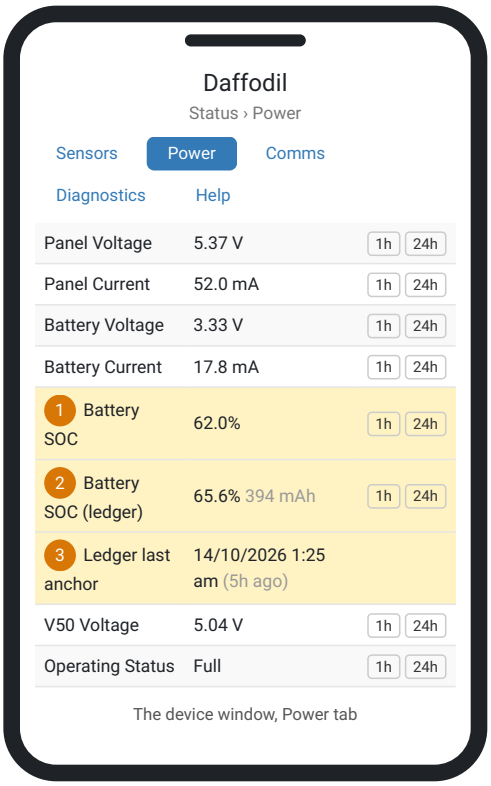
BASICS

The Teleonome page shows how full the battery is as a percentage, the state of charge (SOC). Daffodil's battery is a lithium iron (LiFePO4) cell. Its voltage hardly changes between about a third full and four-fifths full, so the voltage alone can't tell you whether it is half full or three-quarters full.

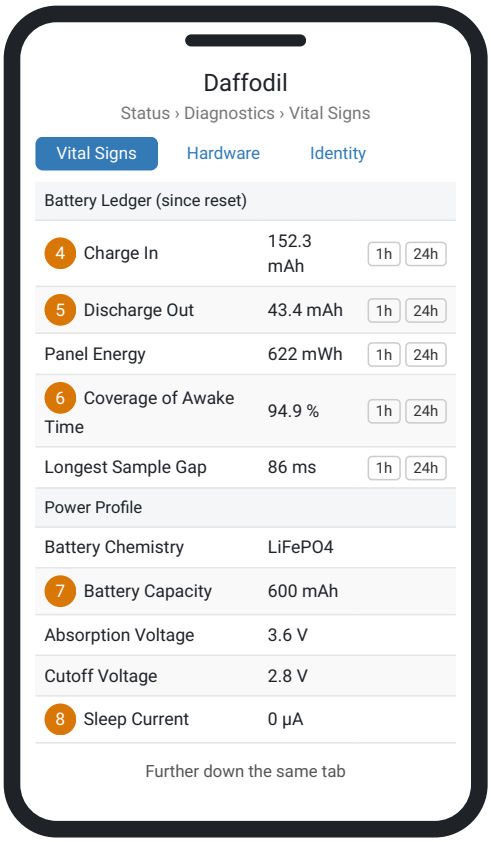
So Daffodil keeps a battery ledger, like a bank statement for its battery. While it is awake it measures the current going into and out of the battery many times a second and adds it up. Each report sends the running totals to Annabelle, which turns them into a percentage. A message that doesn't arrive loses nothing, because the next one carries the totals.

Now and then Annabelle checks the count against the battery voltage and corrects it. These checks are called anchors. They happen when the voltage can be trusted: when the battery is nearly full on a sunny afternoon, and at most once every 6 hours while the unit is resting. A small jump in the percentage at those moments is normal.

What you see on your phone



The device window, Power tab

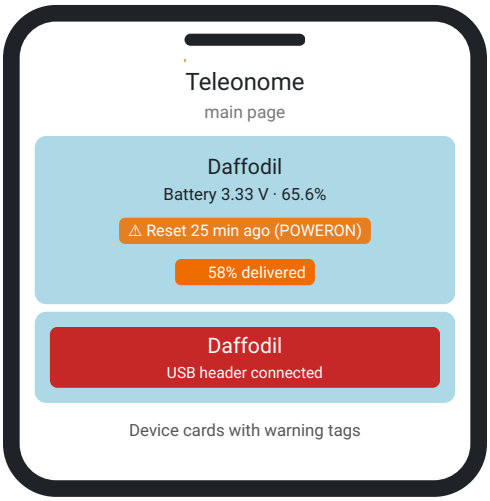


Further down the same tab

#	Row	What it means
1	Battery SOC	The older estimate. It leans on the voltage, so at night it often sits flat (for example at 70%) and then steps.

2	Battery SOC (ledger)	The estimate from the battery ledger, with the charge in mAh (600 mAh = 100%). It moves smoothly as the battery charges and drains. The two figures are shown side by side while they are compared; the ledger figure will replace the older one.
3	Ledger last anchor	When the ledger was last corrected from the voltage, and how long ago.
4 5	Charge In, Discharge Out	Charge that went into and came out of the battery since the unit last restarted.
6	Coverage of Awake Time	How much of the awake time was measured. Around 95% is normal; the first seconds after each wake are not.
7 8	Power Profile	What Daffodil told Annabelle about its battery: type, size, and its current while asleep.

The 1h and 24h buttons draw a graph of that value over the last hour or day. On the main page, the unit's card shows warning tags only while something needs attention:



An orange Reset tag means a normal restart in the last hour; red means a fault restart in the last 24 hours. A % delivered tag means fewer than 80% of the unit's messages are reaching Annabelle. A red name with USB header connected means the programming jumper was left on (chapter 8).

DETAILS

How the ledger is kept, how the percentage is calculated, and how anchors work.

On Daffodil: keeping the ledger

While Daffodil is awake, its second processor core reads the battery and panel current sensors every 70 ms. Each reading is multiplied by the time since the previous one and added to running totals, kept in memory that survives deep sleep:

Total	What is added
Charge In Total (mAh)	battery current while negative (charging)

Discharge Out Total (mAh)	battery current while positive (discharging)
Panel Energy Total (mWh)	panel voltage × panel current
Ledger Measured Seconds	the time actually sampled
Ledger Max Gap (ms)	the longest gap between two samples (should stay near 70)
Slept Seconds Total	time spent asleep (nothing can be measured then)

The adding uses the average of each two neighbouring readings (the trapezoid rule):

```
charge += (I_previous + I_now) ÷ 2 × Δt
```

Before each sleep, Daffodil sends the totals in its Vital Signs message. They return to zero only when the unit really restarts.

The power profile

After a restart and once a day Daffodil also sends its power profile: battery chemistry (LiFePO4), cells in series (1), capacity (600 mAh), full and cut-off voltages (3.6 V and 2.8 V), the current sensors' resolution, and its sleep current. The sleep current is still 0 until it has been measured on the bench, so for now the overnight percentage reads a little high.

On Annabelle: from totals to a percentage

Annabelle's Teleonome runs a task called LedgerSoc every time it receives a new Vital Signs message. It subtracts the previous totals from the new ones and works out the change:

```
change (mAh) =  ΔCharge In × 0.98                    (charging efficiency)
             − ΔDischarge Out × (Δawake ÷ Δmeasured)  (adds the unmeasured first
seconds, up to ×1.5)
             − Δslept (s) × sleep current (µA) ÷ 1000 ÷ 3600

charge = charge + change           (kept between 0 and 600 mAh)
SOC %  = charge ÷ 600 × 100
```

If the reset count changed, the totals started again from zero, so that step adds nothing and the next one carries on.

Example. Between two messages 10 minutes apart at night: Discharge Out went from 38.11 to 38.29 mAh, awake time grew by 12 s of which 11.5 s were measured, and Charge In didn't change. Change = −0.18 × (12 ÷ 11.5) = −0.19 mAh, about 0.03% of the battery.

Anchors

Counting drifts slowly: the unmeasured sleep current, the charging efficiency and small sensor offsets all add up over days. The battery voltage, at rest, gives an independent reading from the chemistry's table:

Voltage (LiFePO4, at rest)	3.45	3.40	3.35	3.30	3.25	3.20	3.10	3.00
SOC	95%	90%	80%	70%	60%	50%	30%	10%

Between 3.20 and 3.35 V the curve is almost flat: 0.01 V is about 2% of charge. So the voltage is a good judge near full and a poor one in the middle. LedgerSoc uses it in three ways:

Anchor	When	Why
Start	First message after the task starts	There is nothing to count from yet.
Reliable zone	3.40 V or more and not charging; at most once an hour	The curve is steep there, so the voltage is accurate. Usually on sunny afternoons.
Resting	3.05–3.40 V while the unit is at rest; at most once every 6 hours	Corrects large drift, at the cost of a few percent of noise.

The size of the jump at an anchor is how far the count had drifted from the voltage. Near full, that is a real measure of the ledger's error; in the flat middle it is partly voltage noise.

A real night, for comparison. Over one night the ledger measured 13.4 mAh used while awake (5-minute reports, plus 90-second reports before dawn). The ledger figure went smoothly from 73.7% to 72.7% by midnight, then a resting anchor at 1:25 am, at 3.29 V, moved it to 68%. The older figure stayed flat at 70% for hours, then stepped down.